

AWS EC2 INSTANCE LAUNCH AND SSH ACCESS REPORT

1. Objective

To launch a virtual server on AWS EC2 and access it securely using SSH, verifying successful connection and deployment.

2. Problem Statement

The task was to create an EC2 instance using the AWS Management Console and ensure successful remote access via SSH. The instance should be properly configured with security settings and should be accessible over the internet.

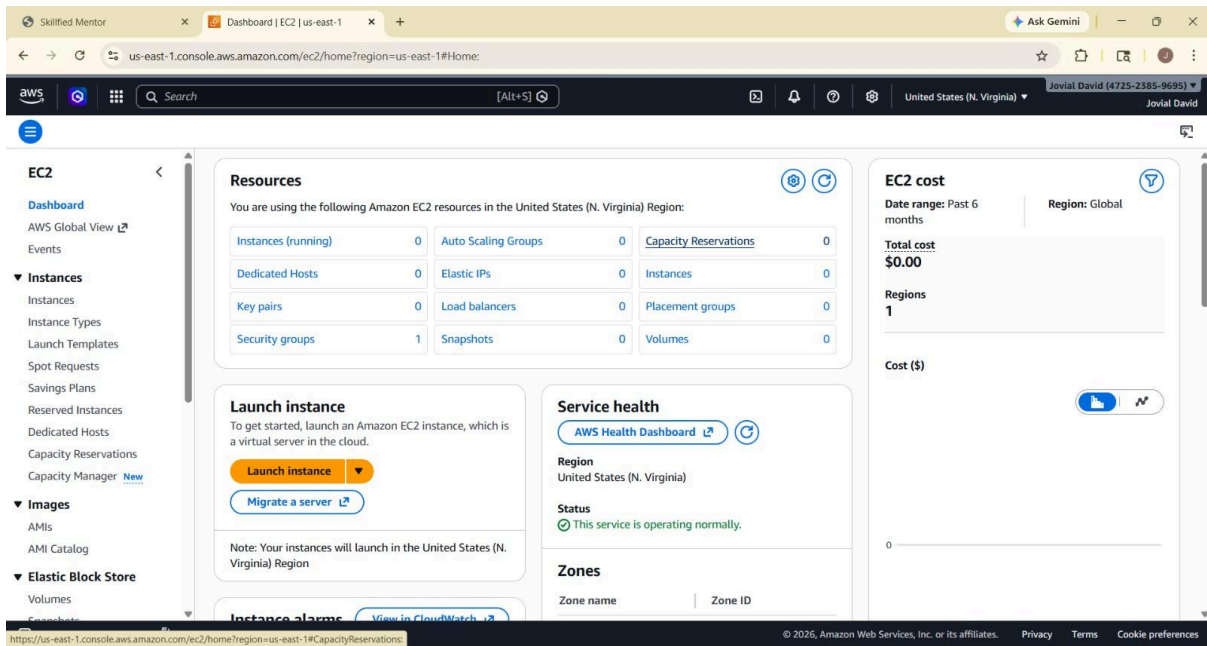
3. Tools and Technologies Used

- AWS Management Console
 - Amazon EC2 (Elastic Compute Cloud)
 - Amazon Linux (AMI)
 - Apache HTTP Server (httpd)
 - SSH (Secure Shell)
 - Web Browser
-

4. Procedure

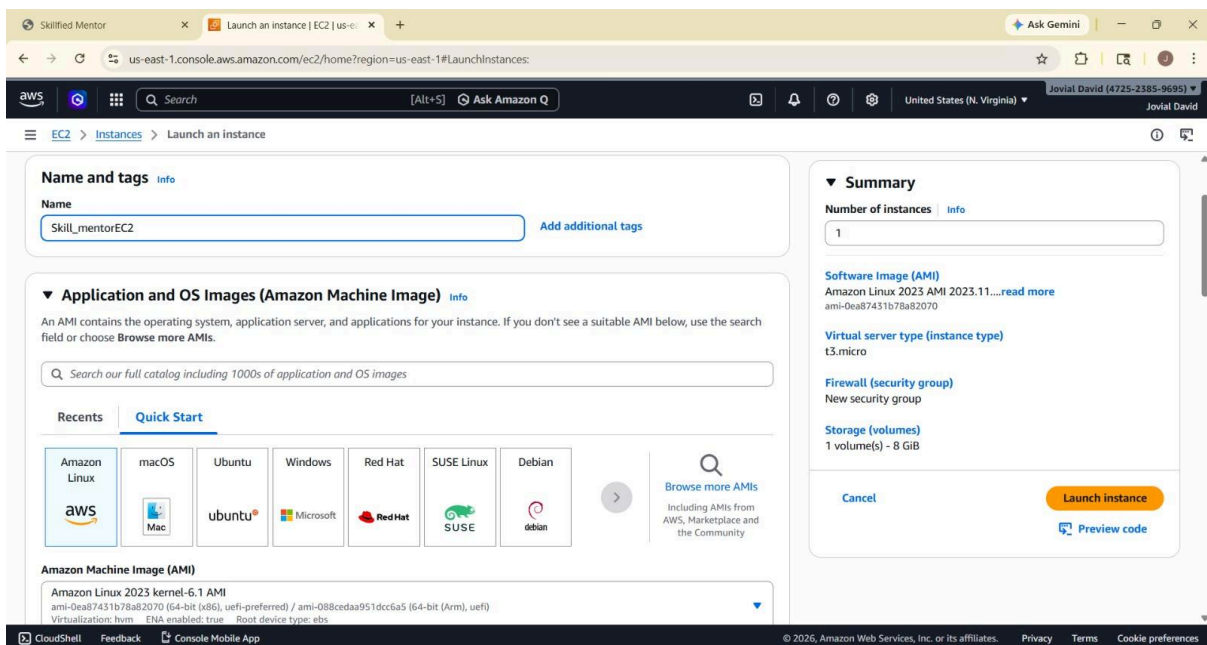
Step 1: Login to AWS Console

Logged into AWS Management Console and navigated to EC2 Dashboard.



Step 2: Launch EC2 Instance

- Selected Amazon Linux AMI
- Chose instance type: t2.micro
- Configured instance settings



Step 3: Configure Key Pair

- Created a new key pair (.pem file)
- Used for secure SSH authentication

The screenshot shows the 'Launch an instance' page in the AWS Management Console, specifically the 'Key pair (login)' step. The 'Instance type' is set to 't2.micro'. The 'Key pair name' is 'my-key', and the 'Create new key pair' checkbox is checked. The 'Network settings' section is partially visible at the bottom. The right-hand 'Summary' panel shows the configuration: 1 instance, Amazon Linux 2023 AMI, t2.micro instance type, New security group, and 1 volume (8 GiB). The 'Launch instance' button is orange and visible at the bottom right.

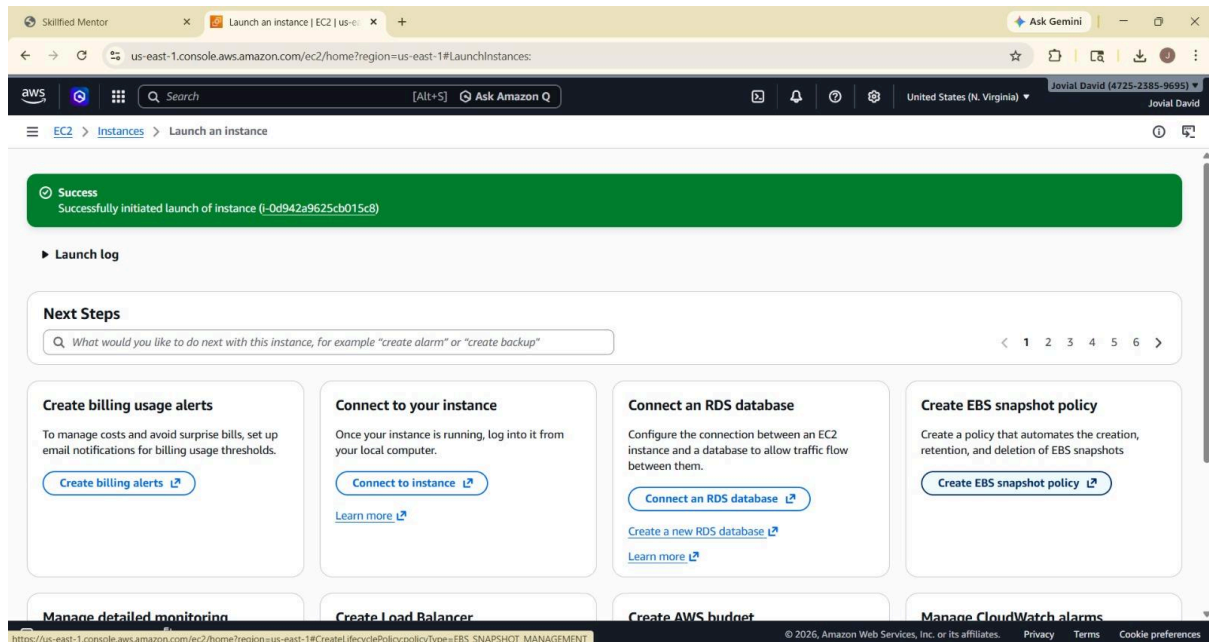
Step 4: Configure Security Group

- Allowed SSH (Port 22)
- Allowed HTTP (Port 80)

The screenshot shows the 'Launch an instance' page in the AWS Management Console, specifically the 'Firewall (security groups)' step. The 'Create security group' radio button is selected. Under 'We'll create a new security group called 'launch-wizard-1' with the following rules:', the 'Allow SSH traffic from' checkbox is checked, and the source is set to 'Anywhere (0.0.0.0/0)'. A warning message states: 'Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.' The right-hand 'Summary' panel shows the configuration: 1 instance, Amazon Linux 2023 AMI, t2.micro instance type, New security group, and 1 volume (8 GiB). The 'Launch instance' button is orange and visible at the bottom right.

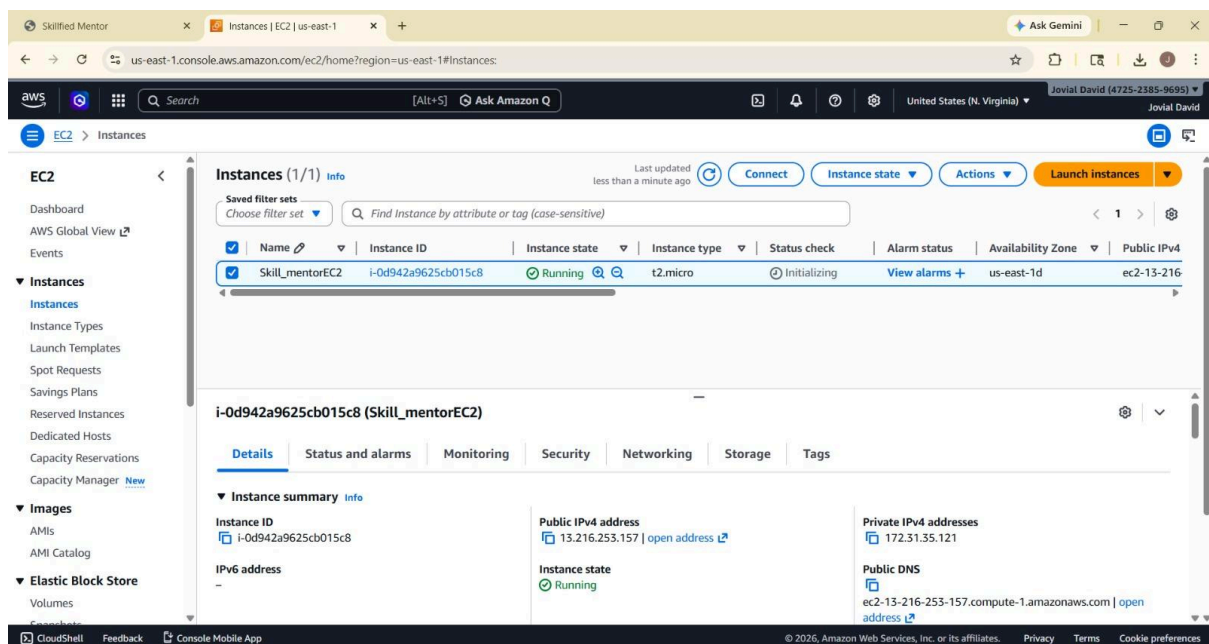
Step 5: Launch and Start Instance

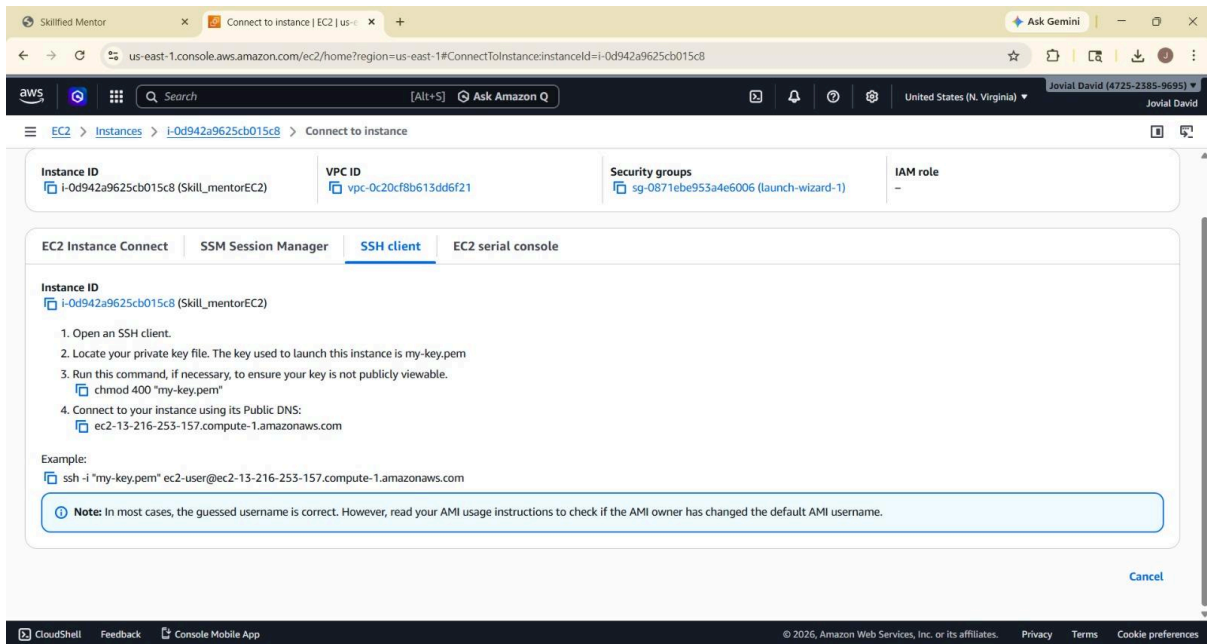
- Instance launched successfully
- Status: Running



Step 6: Connect via SSH

- Used EC2 Instance Connect (browser-based SSH)
- Successfully accessed terminal

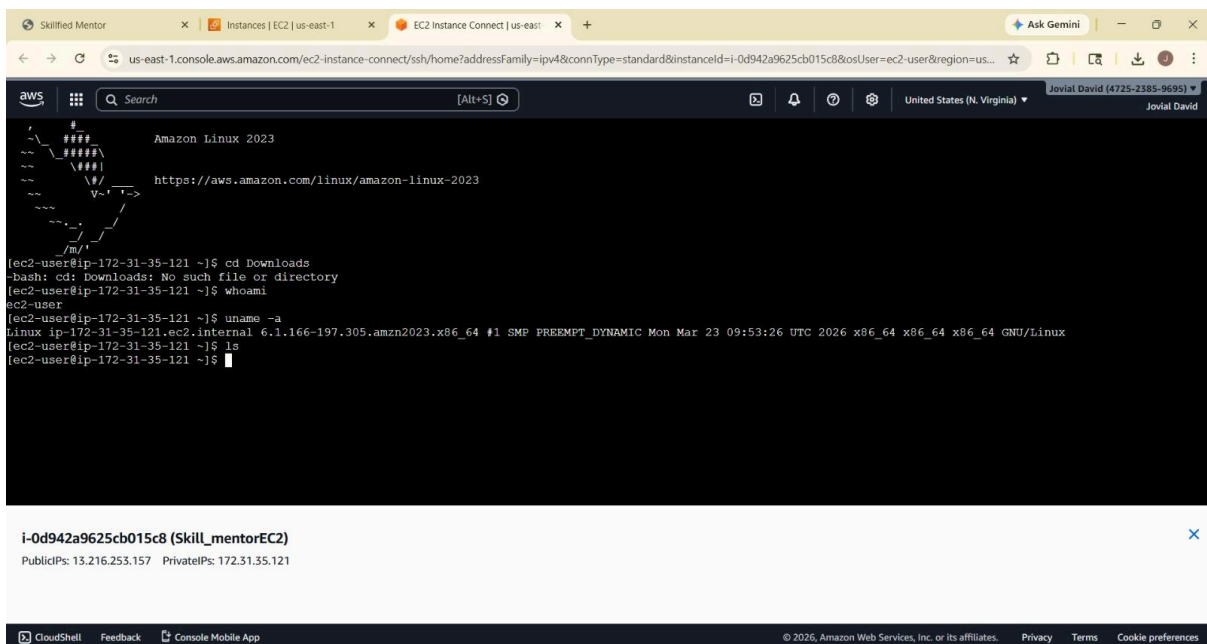




Step 7: Install and Start Apache Server

Commands used:

```
sudo yum install httpd -y
sudo systemctl start httpd
sudo systemctl enable httpd
```



```
aws
Search [Alt+S]
United States (N. Virginia) Jovial David (4725-2385-9695) Jovial David

sudo: systemctl: command not found
[ec2-user@ip-172-31-35-121 ~]$ sudo systemctl start httpd
sudo: systemctl: command not found
[ec2-user@ip-172-31-35-121 ~]$ sudo systemctl start httpd
[ec2-user@ip-172-31-35-121 ~]$
[ec2-user@ip-172-31-35-121 ~]$ sudo systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[ec2-user@ip-172-31-35-121 ~]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Tue 2026-04-14 03:42:51 UTC; 58s ago
     Docs: man:httpd.service(8)
   Main PID: 26520 (httpd)
  Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
    Tasks: 177 (limit: 1120)
   Memory: 13.0M
      CPU: 90ms
   CGroup: /system.slice/httpd.service
           └─26520 /usr/sbin/httpd -DFOREGROUND
             └─26541 /usr/sbin/httpd -DFOREGROUND
               └─26544 /usr/sbin/httpd -DFOREGROUND
                 └─26545 /usr/sbin/httpd -DFOREGROUND
                   └─26546 /usr/sbin/httpd -DFOREGROUND

Apr 14 03:42:51 ip-172-31-35-121.ec2.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Apr 14 03:42:51 ip-172-31-35-121.ec2.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Apr 14 03:42:51 ip-172-31-35-121.ec2.internal httpd[26520]: Server configured, listening on: port 80
[ec2-user@ip-172-31-35-121 ~]$
```

i-0d942a9625cb015c8 (Skill_mentorEC2)

PublicIPs: 13.216.253.157 PrivateIPs: 172.31.35.121

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Step 8: Deploy Web Page

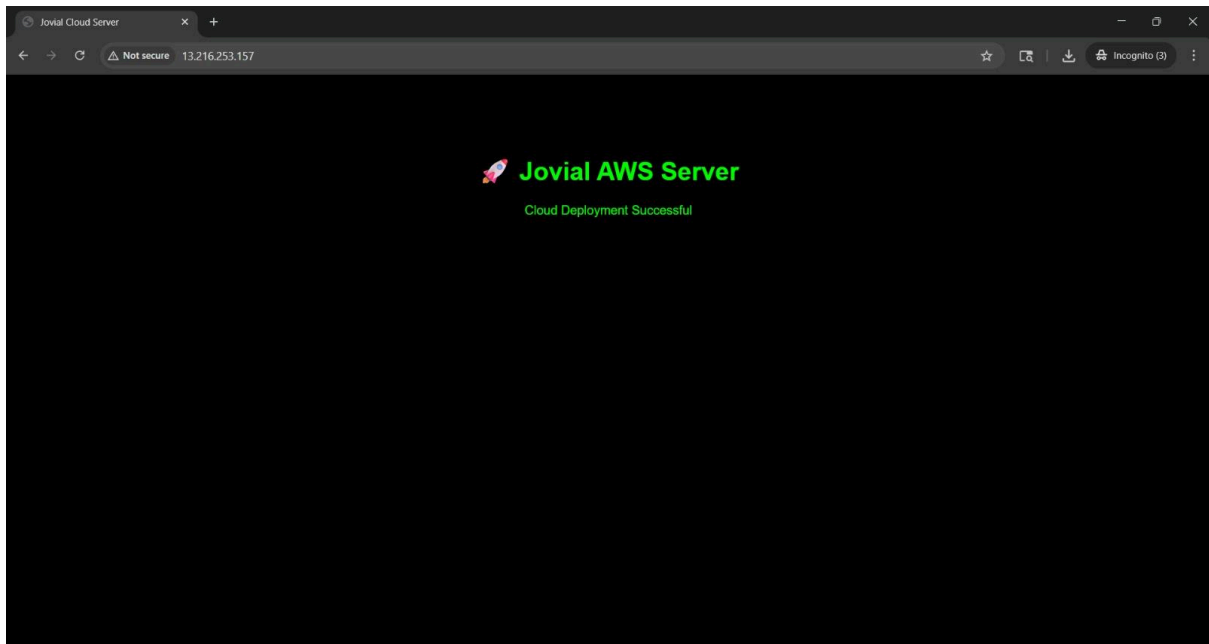
Edited default HTML file:

```
sudo nano /var/www/html/index.html
```

Added custom content and saved.

Step 9: Verify Deployment

Opened the public IP address in browser and confirmed that the custom web page was displayed successfully.



5. Result

A fully functional EC2 instance was successfully launched and accessed via SSH. A web server was deployed using Apache, and a custom web page was hosted and accessed through the public IP.

6. Challenges Faced

- SSH connection failure due to incorrect security group settings
 - Terminal freezing in browser-based SSH
 - Browser caching preventing updates from appearing
-

7. Solutions Implemented

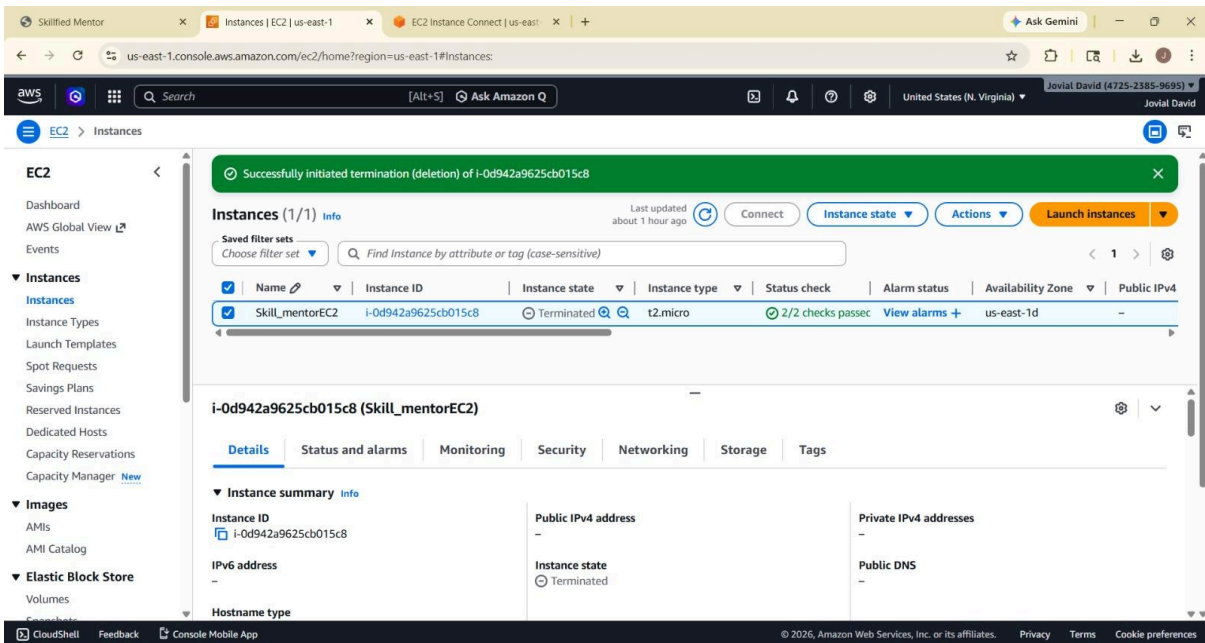
- Updated security group to allow required ports
 - Reconnected and reset SSH session
 - Used force refresh (Ctrl + Shift + R) to clear cache
-

8. Conclusion

The task was successfully completed by launching an EC2 instance, configuring secure access, and deploying a live web server. This practical exercise provided hands-on experience in cloud computing, server management, and troubleshooting real-world issues.

9. Cost Management

All AWS resources (EC2 instance, volumes, and associated services) were terminated after completion to avoid unnecessary billing.



us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#SecurityGroups:

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#SecurityGroups:

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#SecurityGroups:

Security Groups

Security Groups (2)

Find security groups by attribute or tag

	Name	Security group ID	Security group name	VPC ID	Description
☐	-	sg-0871ebe953a4e6006	launch-wizard-1	ypc-0c20cf8b613dd6f21	launch-wizard-1 cre
☐	-	sg-0ef87dd6d5d0b7e95	default	ypc-0c20cf8b613dd6f21	default VPC security

Select a security group

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us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#KeyPairs:

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#KeyPairs:

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#KeyPairs:

Key pairs

Successfully deleted 1 key pair

Find Key Pair by attribute or tag

	Name	Type	Created	Fingerprint	ID
No key pairs to display					

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Skilled Mentor

Elastic IP addresses | EC2 | us-east-1

EC2 Instance Connect | us-east-1

Ask Gemini

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Addresses:

aws

Search

[Alt+S] Ask Amazon Q

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Jovial David (4725-2385-9695)

EC2 > Elastic IP addresses

▼ Images

AMIs

AMI Catalog

▼ Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

▼ Network & Security

Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

▼ Load Balancing

Load Balancers

Target Groups

Trust Stores

▼ Auto Scaling

Auto Scaling Groups

Elastic IP addresses

info

Find elastic IP addresses by attribute or tag

Actions

Allocate Elastic IP address

Name

Allocated IPv4 address

Type

Allocation ID

Reverse DNS record

Associate

No Elastic IP addresses found in this Region

Select an elastic IP address

View IP address usage and recommendations to release unused IPs with [Public IP insights](#)

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